



# 18th edition SQLDay Conference

11-13 May 2026, WROCŁAW + ONLINE

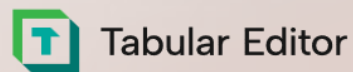
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# Microsoft Fabric: SQL Databases

Kamil Nowinski





Data  
Platform



Microsoft®  
Most Valuable  
Professional



Kamil Nowinski

Microsoft Data Platform **MVP**, **Databricks Champion**  
Cloud Data Architect, Data Strategist, Speaker, Blogger,

Group Manager at [Avanade](#) (UK)

>25 years of experience in IT

Founder of blog [AzurePlayer](#)

GitHub: [#adftools](#), FabricTools and more...

Member of the Data Community PL

Former co-organiser of SQLDay (PL), Data Relay (UK),

volunteer at SQLBits (UK)

### Certificates:

SQL, Data Platform, DevOps, Fabric, Databricks, AI

Moreover: Bicycle, Running, Digital photography, Aikido

[@AzurePlayer.bsky.social](#)

# Blog

- Technical posts
- Various skill level
- Cheat sheets
- Recommended books
- Many useful other links
- Interviews (Podcast)
- YouTube Channel:  
[www.AzurePlayer.net/YouTube](http://www.AzurePlayer.net/YouTube)



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#### Query data in a KQL database in Microsoft Fabric | Lab 12

Kamil Data Geek - Azure explained • 368 views • 2 months ago



#### Get started with Data Activator in Microsoft Fabric | Lab 11

Kamil Data Geek - Azure explained • 705 views • 2 months ago



#### Ingest data with Spark and Microsoft Fabric notebooks | Lab 10

Kamil Data Geek - Azure explained • 498 views • 3 months ago



#### Use real time eventstreams in Microsoft Fabric | Lab 09

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This set of videos promote episodes of "Ask SQL Family" podcast.



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Kamil Data Geek - Azure explained  
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# Slides available:



<https://azureplayer.net/slides>

Jaki jest Twój obecny poziom doświadczenia z SQL Database w Microsoft Fabric?



Poll created by speaker, Kamil Nowinski



Hide poll from attendees

Your answer:

#### Poll Results

11 Answers

Dopiero zaczynam – nigdy nie korzystałem(am) z SQL Database in Fabric



3 / 11

27 %

Słyszałem(am) o tym, ale jeszcze nie próbowałem(am)



3 / 11

27 %

Testowałem(am) na środowisku dev/sandbox



3 / 11

27 %

Mam już wdrożenie lub aktywnie buduję rozwiązanie produkcyjne



2 / 11

18 %

Co najbardziej chcesz wynieść z tej sesji?



Poll created by speaker, Kamil Nowinski



Hide poll from attendees

Your answer:

#### Poll Results

10 Answers

Zrozumieć, czym SQL Database in Fabric różni się od Azure SQL / SQL Server



5 / 10

50 %

Zobaczyć praktyczny przykład wdrożenia krok po kroku



6 / 10

60 %

Dowiedzieć się, jak integrować SQL Database z innymi elementami Fabric



6 / 10

60 %

Poznać best practices dotyczące architektury i zarządzania

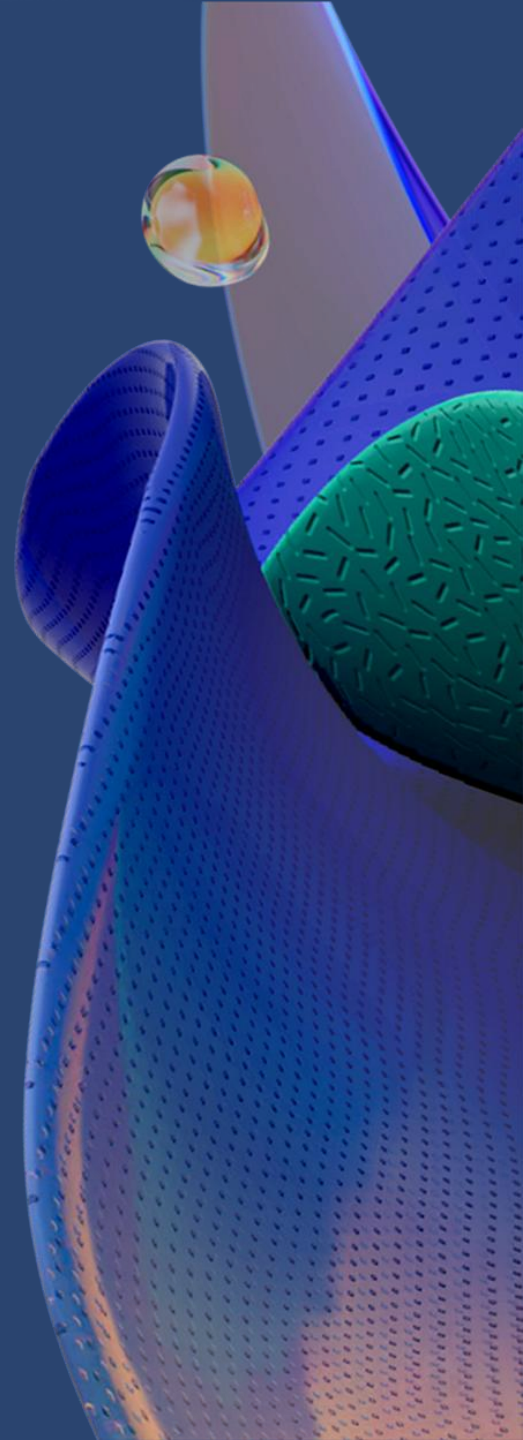


8 / 10

80 %

# Agenda

- [New] Fabric offering: SQL Database
- Simple
- Architecture
- Autonomous & Secure
- Demo
- Summary
- What's next



# Microsoft SQL – Ground to Cloud to Fabric

One SQL for the era of AI

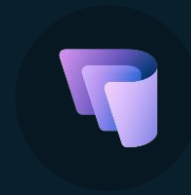
Develop once, deploy anywhere



SQL Server 2025



Azure SQL



SQL database in Fabric

T-SQL

Developers



Engine



Tools



Fabric



AI



Copilots

# Microsoft Fabric

The unified data platform for AI transformation



Data Factory



Analytics



Databases



Real-Time  
Intelligence



IQ



Power BI

Fabric Platform



AI



OneLake



Security & Governance

# Where are you with AI?



**writing  
the code  
in 4 hours**



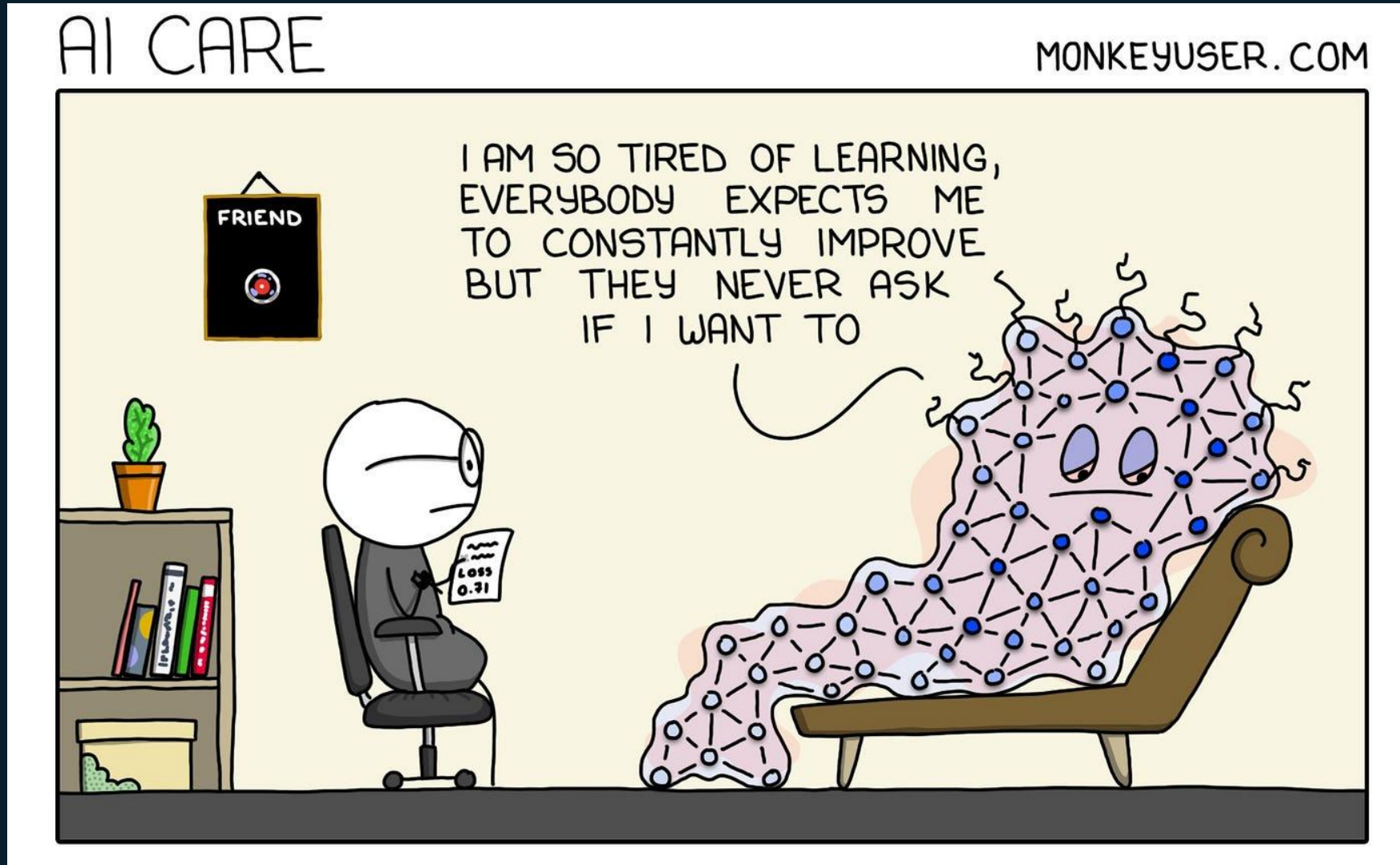
**AI generates  
the code  
in 5 minutes**

**debugging  
the code  
for 10 hours**

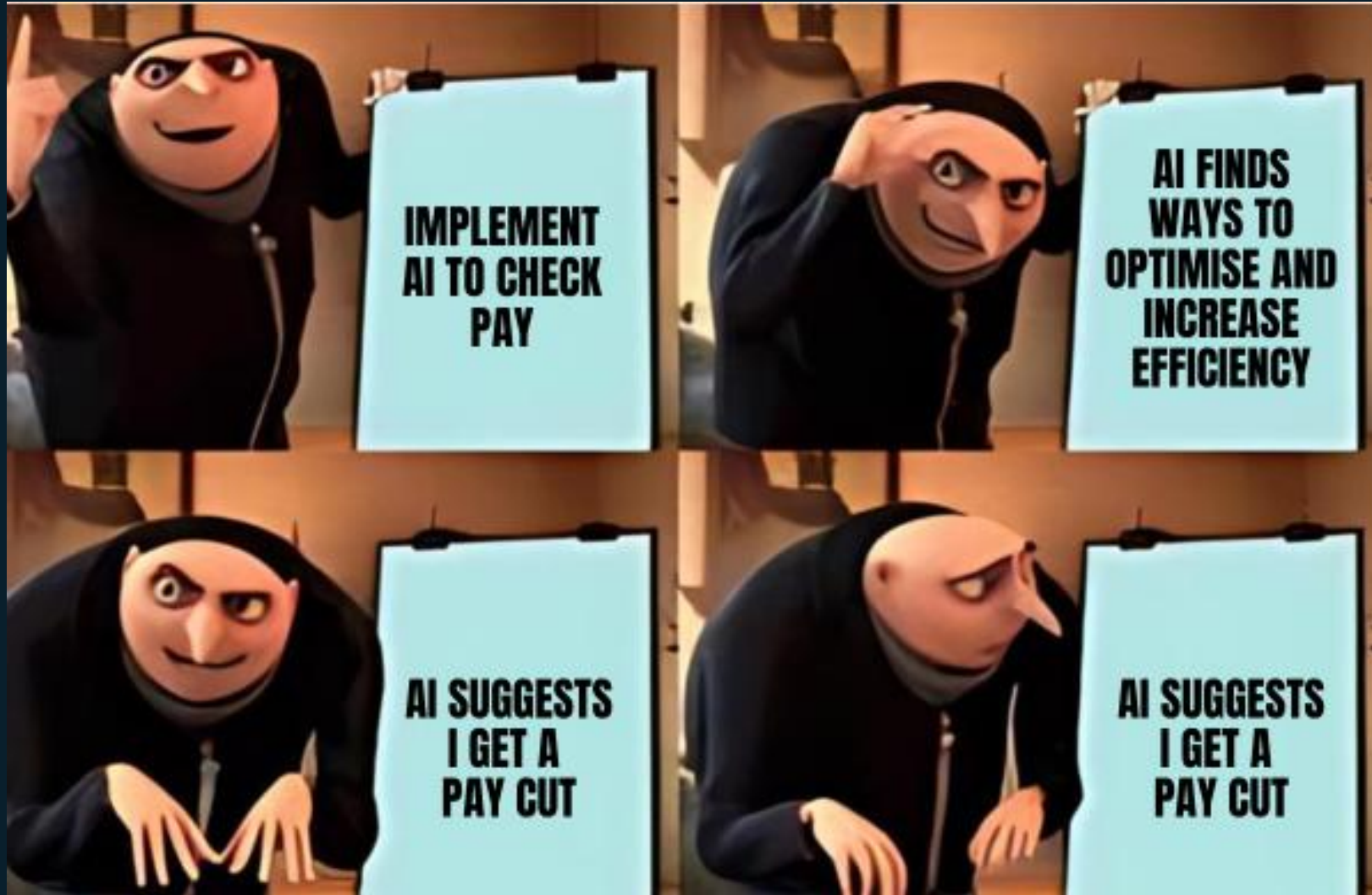
# Where are you with AI?



# Where are you with AI?



# Where are you with AI?



# Microsoft SQL in the cloud

A unified SQL portfolio built on the industry-leading SQL Server engine

SQL Server on Azure  
Virtual Machines



Best for re-hosting and apps requiring OS-level access and control  
Automated manageability features and OS-level access

Infrastructure as a Service

Azure SQL Managed  
Instance



Best for modernizing existing apps  
Offers high compatibility with SQL Server and native VNET support

Platform as a Service

Azure SQL  
Database



Best for building new apps in Azure  
Pre-provisioned or serverless compute and Hyperscale storage to meet demanding workload requirements

SQL Database in  
Fabric



Best for SaaS experience  
OLTP to insights in real-time  
AI-assisted, simplified, autonomous platform  
Deeply integrated with Fabric  
Database source control

Software as a Service



# SQL database in Fabric

Build AI apps faster and easier than ever



Simple



Autonomous  
and secure



Optimized  
for AI

Built upon the familiar Microsoft SQL engine, in Microsoft Fabric

Simple

# Focus on building your app

A simple, integrated developer experience



Easy provisioning



AI assistance with Copilot



T-SQL support



Unified billing with Fabric capacity model

The screenshot displays the Microsoft Fabric interface. At the top, there are navigation tabs: Home, Replication, Security, and Help. Below these are icons for Get data, New Query, Templates, Open in, and New API. The main interface is divided into two panels. The left panel, titled 'Explorer', shows a tree view of the database structure. It includes a search bar and a list of items: CustomerRiskDB (expanded), dbo (expanded), Tables (expanded), Customers, RiskScores (selected), Transactions, Views, Stored Procedures, Functions, External Tables, Queries (expanded), My queries, and Shared queries. The right panel, titled 'Data preview - RiskScores', shows a table with columns: RiskScoreId, CustomerId, and Trail. The table contains 17 rows of data.

|    | RiskScoreId | CustomerId | Trail |
|----|-------------|------------|-------|
| 1  | 1           | 5          | 358   |
| 2  | 2           | 1          | 98    |
| 3  | 3           | 4          | 107   |
| 4  | 4           | 1          | 135   |
| 5  | 5           | 9          | 10    |
| 6  | 6           | 9          | 47    |
| 7  | 7           | 5          | 288   |
| 8  | 8           | 5          | 380   |
| 9  | 9           | 6          | 126   |
| 10 | 10          | 6          | 465   |
| 11 | 11          | 8          | 416   |
| 12 | 12          | 6          | 324   |
| 13 | 13          | 2          | 301   |
| 14 | 14          | 1          | 499   |
| 15 | 15          | 8          | 356   |
| 16 | 16          | 4          | 274   |
| 17 | 17          | 8          | 383   |

# OneLake Mirroring

Near real-time replication to OneLake by default, seamless integrations with the rest of Fabric



Unified operational and analytics



Data replicated to open-source Delta



Shortcuts and SQL analytics endpoint to query data without copying it

|                                                                                     | Name        | Status | Git status                                                                                 | Type                   |
|-------------------------------------------------------------------------------------|-------------|--------|--------------------------------------------------------------------------------------------|------------------------|
|  | banking_app |        |  Synced | SQL database           |
|  | banking_app |        | —                                                                                          | SQL analytics endpoint |



SQL analytics endpoint ▾



SQL Database

Define, query, and update data



SQL analytics endpoint

Query data using SQL

# Application lifecycle management

Every database under source control



Source control integration in the Fabric web interface (GitHub and Azure DevOps)



Database code editing in developer tooling



SqlPackage command line tool

The screenshot displays the Source control interface in Azure Data Studio. At the top, there are navigation links: 'View deployment pipeline', 'Create app', 'Manage access', and 'Workspace settings'. Below these, a 'Source control' tab is active, showing a list of tasks. A search bar 'Filter by keyword' and a 'Filter' dropdown are also present. The 'Changes' section is highlighted, showing a list of items with checkboxes and status indicators. A 'Commit' button is visible at the bottom of the changes list.

| Task | Owner      | Refreshed   | Ne ref |
|------|------------|-------------|--------|
| —    | Anna Ho... | —           | —      |
| —    | Anna Ho... | —           | —      |
| —    | Contoso... | 7/22/202... | N/A    |
| —    | Anna Ho... | —           | —      |
| —    | Anna Ho... | —           | —      |
| —    | Contoso... | 10/25/20... | N/A    |
| —    | Anna Ho... | —           | —      |
| —    | Anna Ho... | —           | —      |
| —    | Anna Ho... | —           | —      |

**Source control**

Current branch: **planelanding**

**Changes** 8 Updates

- ☐ Contoso PDFs RAG +
- ☐ RAG-with-resumes +
- ☒ api1 %
- ☒ ContosoDB %
- ☒ dev %
- ☒ Notebook 1 -

**Commit** **Undo**

# Migration from SQL to Fabric, simplified

Announcing public preview



Guided migration experience



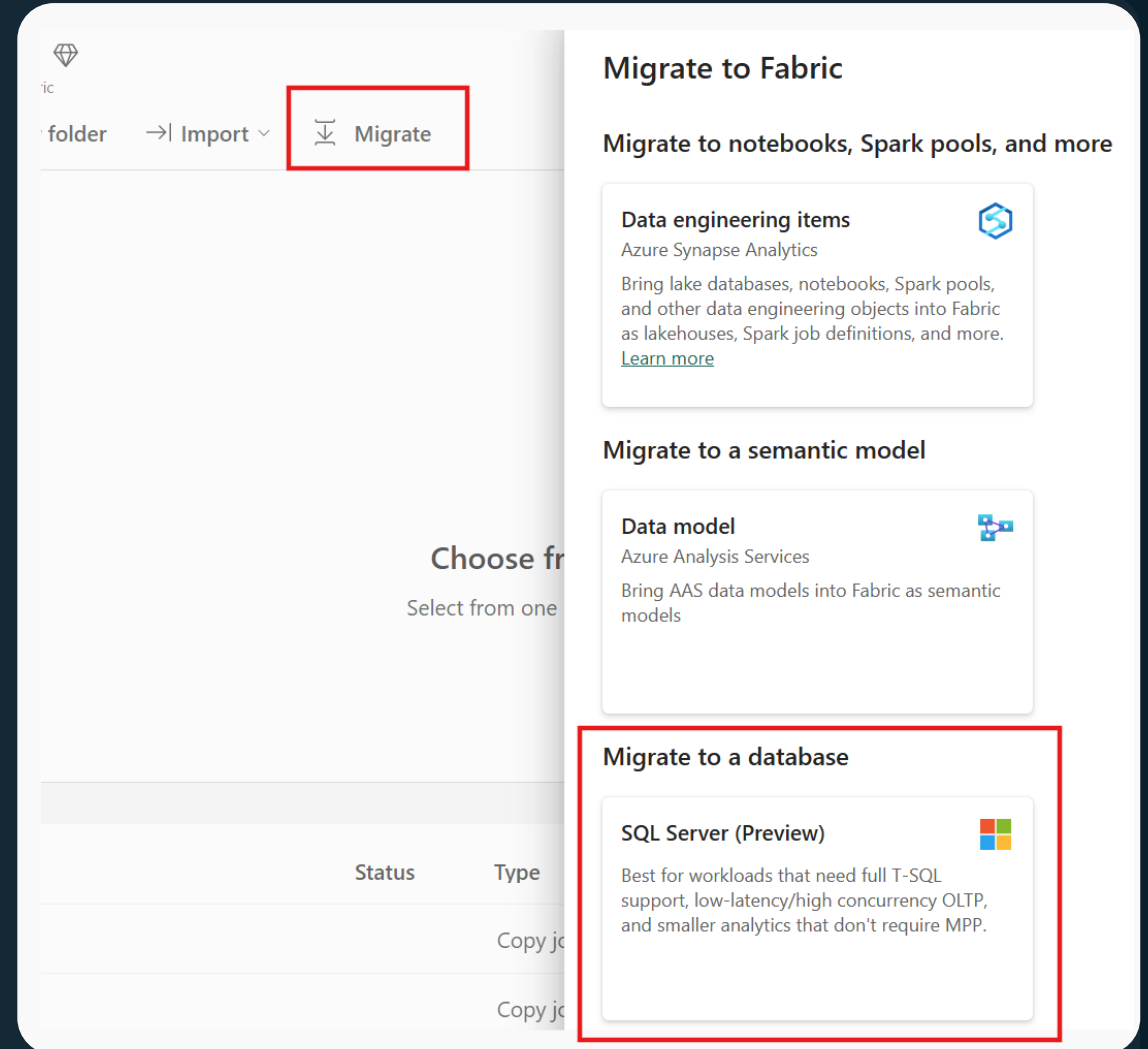
Schema migration with Copilot-assisted error resolution



Data Migration using Fabric Copy Jobs



Faster time to value



# What's New: Simple

- ✓ Migration experience in Fabric\*
- ✓ Point-in-time restore for dropped databases\*
- ✓ List deleted databases by workspace\*
- ✓ Pre/post deployment actions\*
- ✓ Allow custom table selection for mirroring\*
- ✓ Enable SQL projects to open in SSMS\*

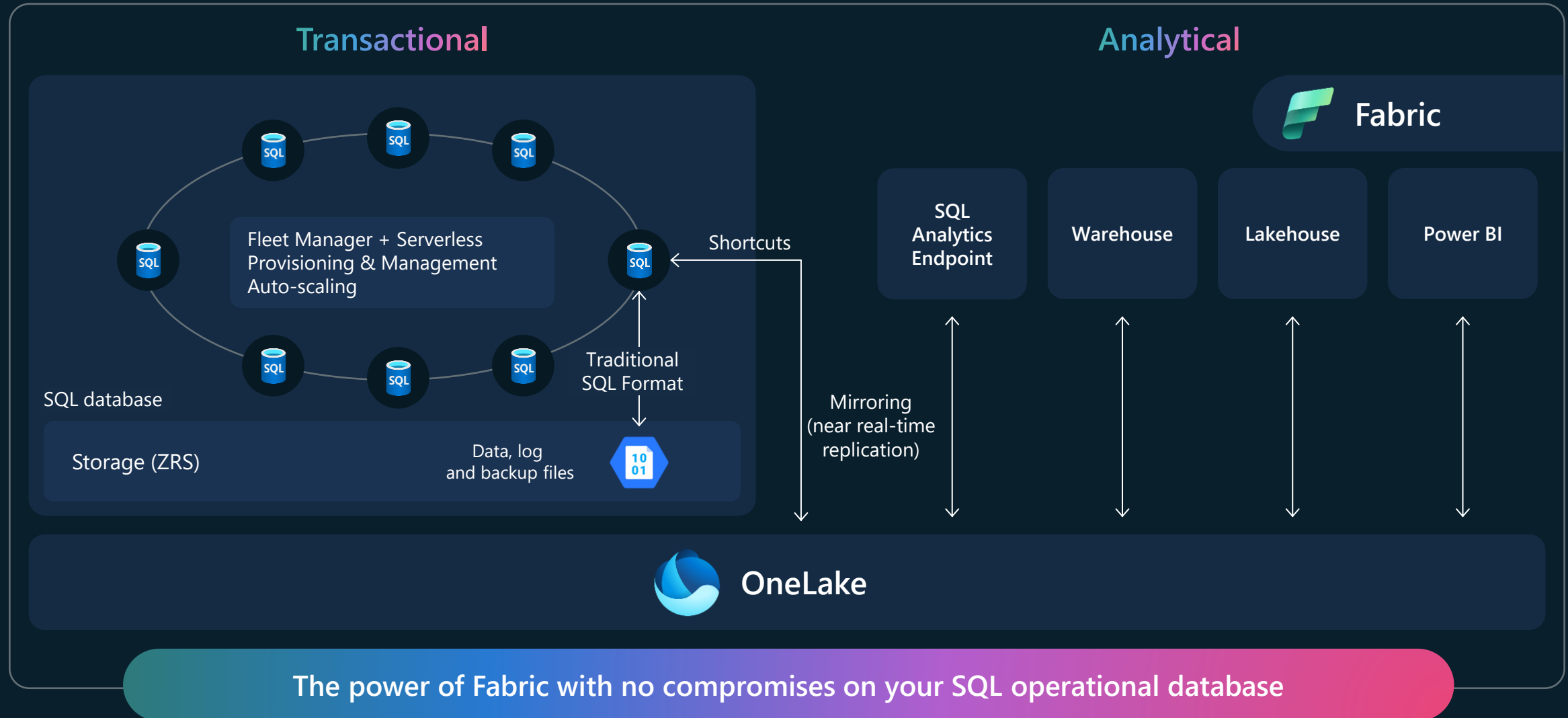
All items Generally Available unless marked:

\*Public Preview

\*\*Private Preview

# Architecture

# SQL database in Fabric Architecture



Demo



# SQL Database

# Top Use Cases and Scenarios



Operational data  
store (ODS)



Event-driven  
apps



Line of business  
Power apps +  
Translytical



Meta-driven  
pipelines



Vector and AI  
app dev



Reverse ETL



Power BI  
writebacks



Reference  
data store

Autonomous  
and secure

# Autonomous capabilities manage your workloads



## Challenge

Making the right choices and managing databases is complex



## Solution

SQL database in Fabric enables you to get the best security, availability, and performance configurations by default



## Autonomous



Auto-HA with Zone Redundancy

Auto-scaling compute and storage

Auto-replicated to OneLake (no ETL)

Automatic Index Compaction (preview), Auto-indexing

Improved Intelligent Query Processing enhancements

Auto-backups, point-in-time restore available

Auto-patching

# What's New: SaaS by default, PaaS configurable

- ✓ All collations support\*
- ✓ Configurable Max vCore\*
- ✓ Full text search support
- ✓ Support for ALTER DATABASE\*
- ✓ GeoDR with failover groups\*\*

## Compute (preview)

Manage database performance and capacity usage by configuring this database's resource limits. [Learn more](#) 

### Max vCore limit

4 vCores



Changing this setting may take a few minutes and result in temporary connectivity disruptions to the database.

Save

Discard

All items Generally Available unless marked:

\*Public Preview

\*\*Private Preview

# Secure and governed by default

## Authentication



Microsoft Entra authentication only

- Central identity management
- Passwordless authentication

## Authorization



- Workspace roles and item permissions for easy management
- Proven SQL native features for granular access control

## Network security



- Private link-ready
- All data encrypted in transit – TLS 1.2, 1.3

## Data encryption



- All data encrypted at rest
- Customer Managed Key support

## Auditing



Smart defaults and customizable

## Central governance



- Microsoft Purview sensitivity labels
- Centrally-managed protection policies

# What's New: Security & Compliance

- ✓ SQL Audit
- ✓ Customer-Managed Keys
- ✓ Dynamic Data Masking

All items Generally Available unless marked:

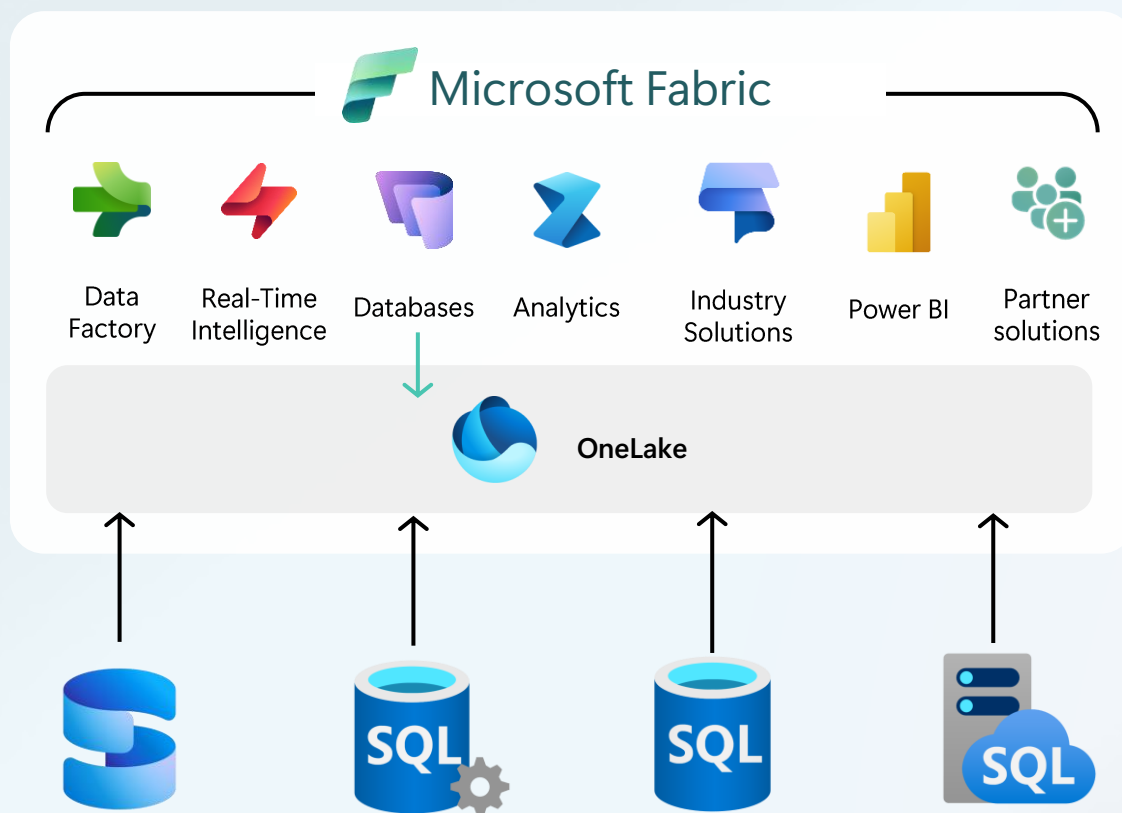
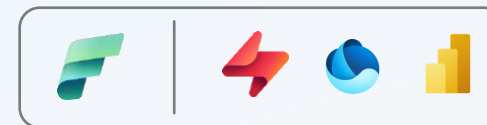
\*Public Preview

\*\*Private Preview

# Mirroring

# Integrated with Microsoft Fabric

Integrate your operational data with a unified data platform



Mirroring replicates databases to Fabric with **zero ETL**

Data is replicated into One Lake and **kept up-to-date in near real-time**

Mirroring **protects operational databases** from **analytical queries**

Compute replication is included with your Fabric capacity **for no cost**.

**Free Mirroring Storage** for Replicas tiered to Fabric Capacity.

# Unified data culture

Transactional

Analytical

# Unified data culture

Translytical

# Summary

# Azure SQL Database <VS> Fabric SQL Database

| Feature                                  | Azure SQL Database       | Fabric SQL Database                               |
|------------------------------------------|--------------------------|---------------------------------------------------|
| Always Encrypted<br>Application Roles    | Yes                      | No                                                |
| Change Data Capture (CDC)                | Yes (Tier S3 and above)  | No                                                |
| Columnstore indexes                      | Yes                      | Yes (with limitations)                            |
| Cross-database (three-part name queries) | No (see Elastic queries) | Yes (via SQL analytics endpoint)                  |
| EXECUTE AS                               | Yes                      | No                                                |
| Server-level roles                       | Yes                      | No                                                |
| SQL Server Agent                         | No (see Elastic jobs)    | No – use Factory Pipelines or Apache Airflow jobs |
| TDE (Transparent Data Encryption)        | Yes                      | No (see Storage Encryption)                       |

# Key Takeaways

- ✓ **Unified Fabric platform** - natively integrated with OneLake, Power BI, pipelines, and Spark with no data movement required.
- ✓ **Automatic mirroring to OneLake** - data is continuously replicated to Delta Parquet format, making it instantly available to all Fabric workloads without ETL.
- ✓ **Fully managed & serverless** - no infrastructure to provision or manage; auto-scaling, high availability, and backups are built in.
- ✓ **T-SQL compatibility** - uses the same SQL Server/Azure SQL engine, so existing skills, queries, and tooling (SSMS, dbt, etc.) work without changes.
- ✓ **Consumption-based pricing** - billed through Fabric capacity with no idle instance costs; the database pauses automatically when not in use.

What's next



# Roadmap: SQL database in Fabric



## Simple

- Data Virtualization: Shortcuts to Fabric OneLake
- Copilot: Agent mode, skills
- Migration from SQL Server improvements
- Schema designer



## Autonomous + secure

- Workspace Level Private Link
- Expanded vCore controls
- Higher memory per core
- 128 TB max size

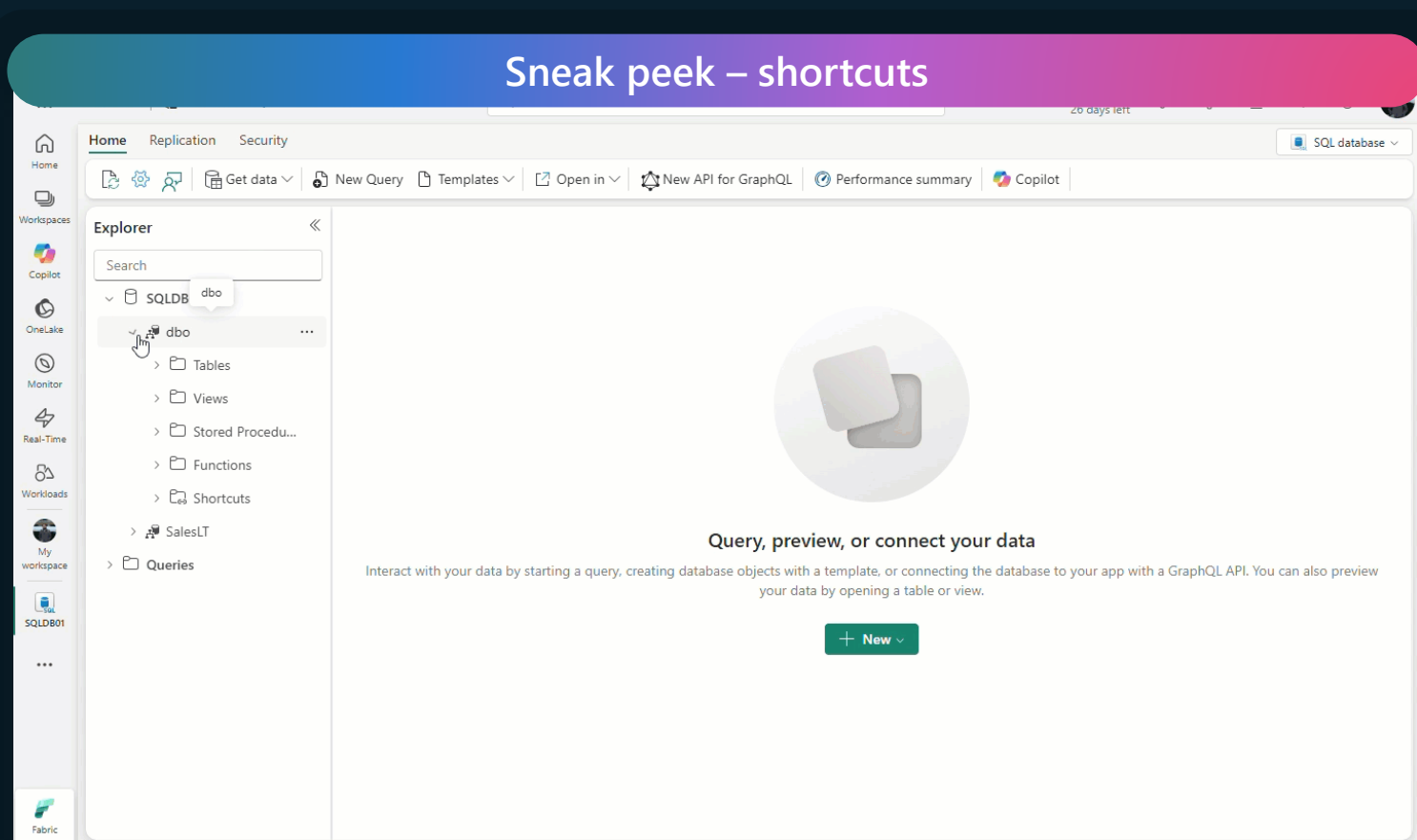


## Optimized for AI

- Database Agents and Database Hub
- Change Event Streaming
- Vector index improvements

[aka.ms/fabricsqlroadmap](https://aka.ms/fabricsqlroadmap)

## Sneak peek – shortcuts



# What's new in Microsoft SQL @SQLCon

## Migrate and Modernize

### Assessment + Migration

- Azure SQL VM migration support with Arc
- SSMA Sybase to SQL with Copilot
- Fabric migration experience to Fabric SQL Database\*
- Register and inventory multiple instances and databases\*
- Azure SQL Managed Instance Link
  - Multi-DB migration\*\*
  - Simplified MI Link creation experience
- Change Event Streaming for SQL Managed Instance
- Private subnets support

### Linux

- Bulk admin role\*
- Adutil packages for Ubuntu 24.04 and RHEL 10
- SQL Server 2025 for Ubuntu 24.04 and RHEL 10
- DMV parity
- Pacemaker HA persistent agent service\*

## Cloud-Native

### Monitoring

- Access performance data via ADX, at-scale monitoring\*
- Resource-specific performance dashboard
- New Query Editor experience in Azure\*

### AI

- SQL MCP Server via Data API builder\*
- DML on VECTOR INDEXed tables\*
- Iterative Filtering for Vector Indexes with TOP APPROXIMATE\*
- Improved CREATE VECTOR INDEX performance\*

### Security & Enterprise

- TDE Versionless key support\*
- Entra principals for login creation
- Private subnet support
- Terraform SQL API coverage

### Performance

- 160 and 192 v-Core support\*
- Load balancing for named replicas\*\*
- Automatic index compaction\*

## Unify your Data Platform

### SQL database in Fabric enterprise readiness

- Migration to SQL databases\*
- Configure database-level v-Core limits\*
- Custom table selection for mirroring\*
- List and restore deleted database\*
- All collations support\*
- Full text search support\*
- Customer-Managed Keys
- Auditing
- Dynamic Data Masking
- Perf dashboard/Workspace Monitoring\*\*
- Azure SQL Database ALTER DATABASE parity\*
- Open SQL Project from Fabric in SSMS\*
- Pre/post update/deployment actions\*

### Mirroring

- Support for mirroring >500 tables

## Tools + SDKs

### VS Code MSSQL extension

- Application quickstart with GitHub Copilot\*, including design schema and data API
- Query Profiler\*
- Edit Data
- DACPAC Import/Export
- SQL Notebooks\*
- Expanded SQL Project item templates
- Fabric SQL browse and create

### SSMS 22

- GitHub Copilot
- Code completions and DB instructions
- SQL Projects\*
- Export to JSON, XML, Excel
- Group by schema

### Drivers & SDKs

- Python Bulk Copy support
- PHP 8.4 + 8.5
- SQLClient 7.0
- JDBC 13.4
- Django 6.0

# aka.ms/mssqldecks

Learn more about  
SQL database in  
Fabric



SQL database in Fabric: The unified database for AI apps and analytics

## Key resources about SQL database in Fabric

- Learn: [aka.ms/FabricSQLdatabase](https://aka.ms/FabricSQLdatabase)
- Blog: [aka.ms/FabSQLConUSBlog](https://aka.ms/FabSQLConUSBlog)

## Explore tutorials & Build AI apps

- Tutorial: [aka.ms/SQLFabric-tutorial](https://aka.ms/SQLFabric-tutorial)
- Hands-on lab: [aka.ms/SQLinFabricAILab](https://aka.ms/SQLinFabricAILab)
- Build Agentic App: [aka.ms/AgenticAppFabric](https://aka.ms/AgenticAppFabric)

## Discover more from our YouTube channel

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# Resources

- Explore [Documentation](#) & [End-to-end tutorials](#)
  - [Data Exposed episodes](#)
  - [Microsoft Mechanics episode](#)
  - [Join the community](#) for a free live learning series starting
  - Get the latest product news here on the [Fabric updates blog](#).
- 
- [My Fabric Labs series](#) (YouTube playlist)
  - [Microsoft Fabric Community](#) (My YouTube playlist)
  - [My GitHub-Fabric list](#)
  - [FabricTools](#) – PowerShell module

# Thank you!



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**Kamil Nowinski**

Microsoft Data Platform MVP, Databricks Champion  
Analytics Architect, Azure DevOps Engineer Expert



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